



UNIVERSITY OF PATRAS
COMPUTER ENGINEERING AND INFORMATICS DEPARTMENT
MSc PROGRAM *INFORMATICS OF LIFE SCIENCES*

MSc topics 2021-2022
Georgios Drakopoulos

1. GNNs for mining biological graphs

Recently modeling and mining biological graphs has garnered intense research interest. Graph Neural Networks (GNNs) is an emerging technique for processing complex biological data which are naturally described in two or more dimensions and their interactions. Under this model genes and proteins serve as the vertices of the graph, whereas any known associations form the edges. Once the graphical model has been formed, vertex classification and edge prediction tasks can be executed in order to obtain latent information from the graph. In this way, missing interactions in sparse, scale-free biomolecular networks can be inferred.